

Design and Analysis of FDM Prototypes for Gold-198 Nanoparticles.

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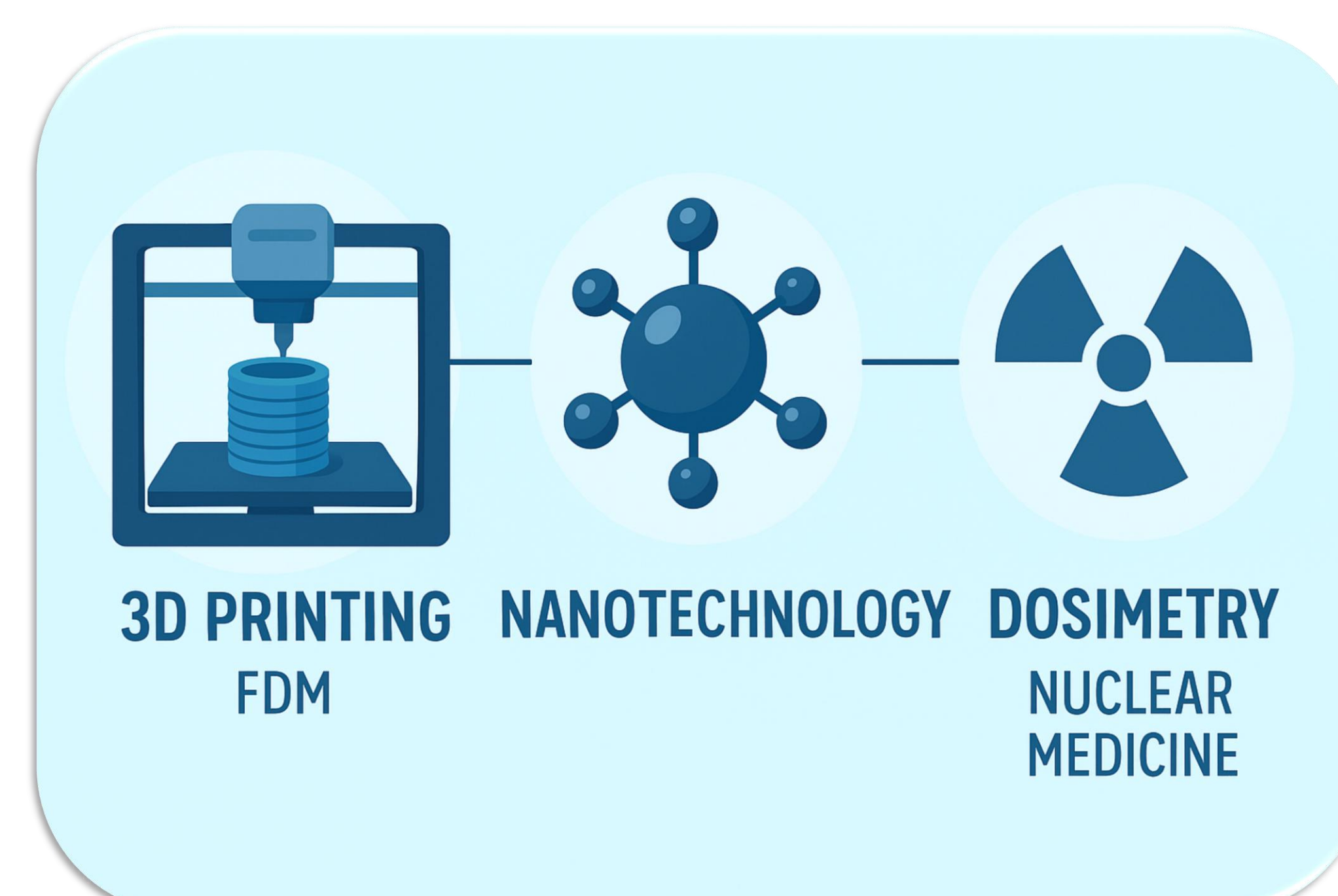
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INTRODUCTION

Additive Manufacturing (3D Printing): A key technology of Industry 4.0, enabling the fabrication of customized, complex, and high-precision structures through a layer-by-layer deposition process, while reducing material waste.

Fused Deposition Modeling (FDM): The most widely used AM technique due to its low cost, accessibility, and versatility in processing thermoplastic filaments. Particularly relevant in medical physics for prototyping experimental systems, dosimetry studies, and supports for radioactive sources



MATERIALS AND METHODS

Synthesis of ¹⁹⁸AuNPs

- Gold dissolved (HCl:HNO₃), stabilized with gum Arabic.
- Reduction: sodium citrate + NaOH → AuNPs (wine-red).
- Neutron activation of ¹⁹⁷Au → ¹⁹⁸AuNPs (T_{1/2} = 2.7 d, 16.65 MBq).

FXG Dosimeter

Gelatin (5%) + H₂SO₄, NaCl, ferrous ammonium sulfate, xylenol orange. Homogenized, stored 12 h at 4°C.

Prototype Design & 3D Printing

CAD model: cylinder (40 mm × 23 mm, wall 1 mm).

Materials: PLA 220°C | ABS 270°C | PETG 250°C | ASA 260°C.

Experimental Setup

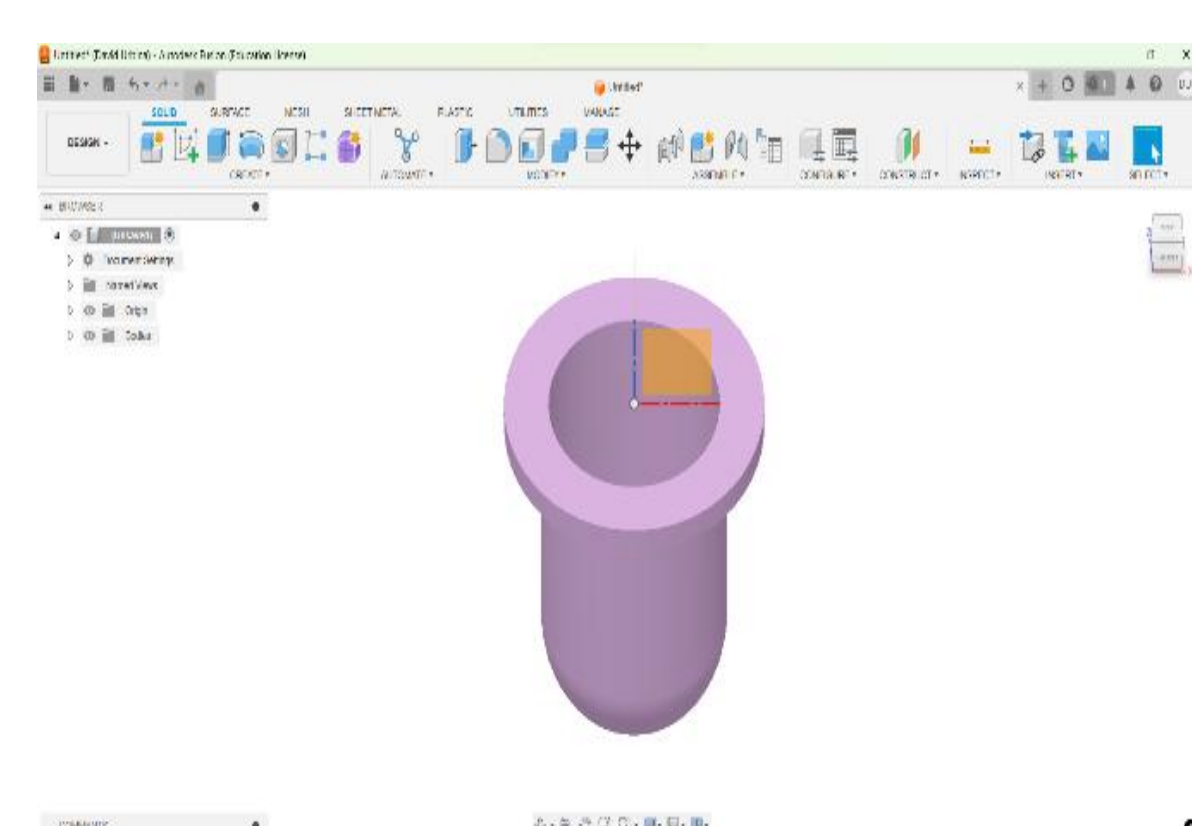
1 mL ¹⁹⁸AuNPs inside printed cylinders

Immersed in FXG, 24 h (dark, refrigerated)

Analyses

OCT → dose distribution (optical density)

FTIR (ATR mode, 4000–550 cm⁻¹, 32 scans) → polymer stability



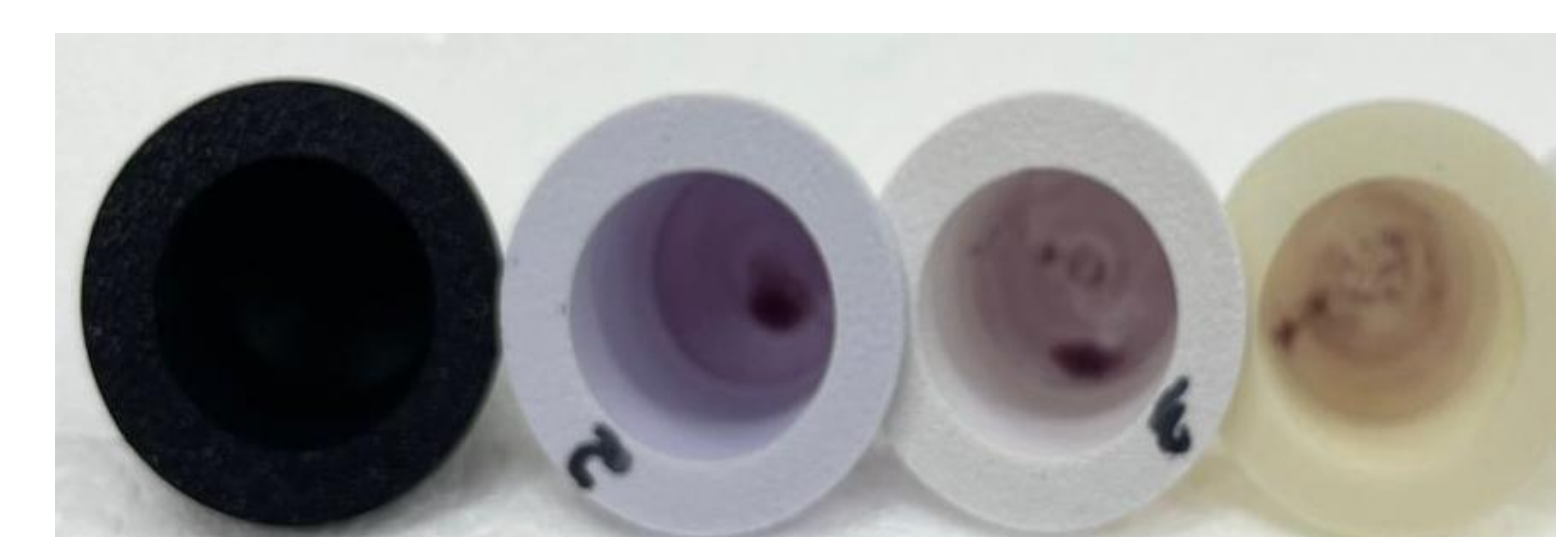
Fusion 360 software.



Bambu Lab X1 Carbon.



FXG with 3D Prototype Supports filled with AuNPs



The 3D Prototype supports after 24hs

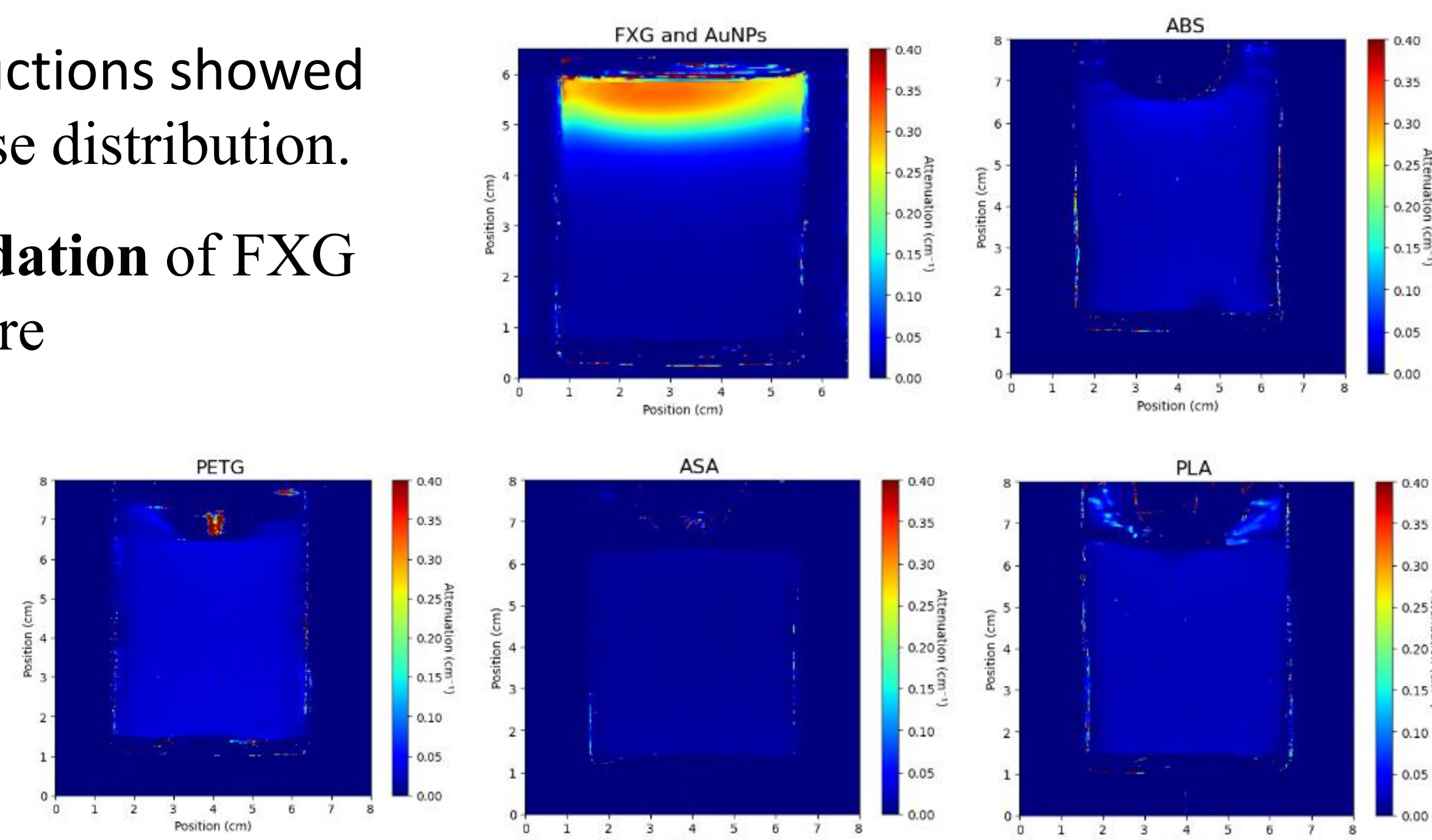


The AuNPs in the different 3D Prototype supports.

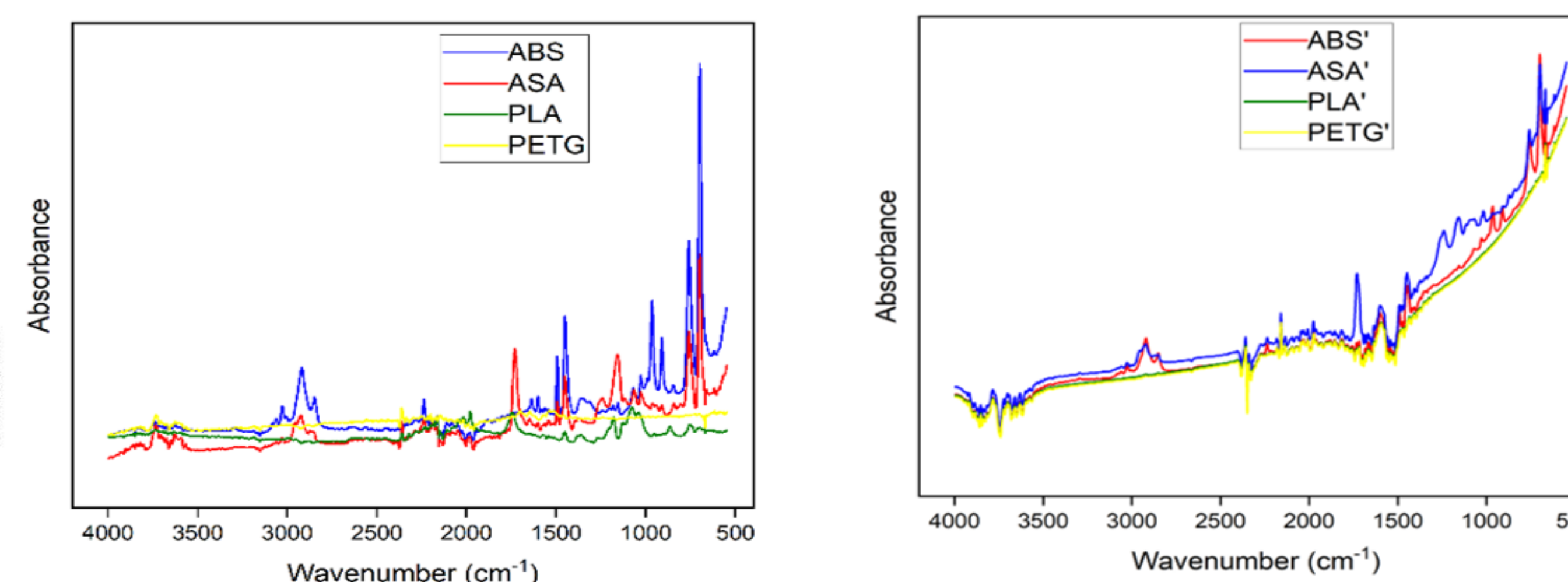
RESULTS

3D OCT reconstructions showed homogeneous dose distribution.

No visual degradation of FXG after 24 h exposure



As-printed polymers displayed **characteristic absorption bands** (ABS/ASA: nitrile ~2230 cm⁻¹; PLA: C=O ~1745 cm⁻¹; PETG: C=O ~1725 cm⁻¹).



(a)FTIR spectra for prototypes in their as-printed condition
(b)FTIR spectra for 3D prototypes support after 24hs of contact with FXG dosimeter and AuNP.

Optical attenuation maps comparing FXG exposed to AuNPs without supports and with ABS, ASA, PETG, and PLA 3D prototypes supports.

CONCLUSIONS

- FTIR analysis confirmed that the polymers retained their chemical integrity after 24 hours of exposure, with spectral variations attributable to surface adsorption of the gel rather than bulk chemical reactions.
- OCT and dose distribution analyses further indicate that the high-energy β^- emission of ¹⁹⁸Au results in localized superficial dose deposition, effectively contained by the polymeric supports.

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